

B. Continuous distributions (contd)

2. r.v. $X \sim \text{Normal}(\mu, \sigma^2)$ with pdf

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp. -\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2, \quad x \in \mathbb{R}; \quad \mu \in \mathbb{R}, \sigma > 0$$

with mgf $M_X(t) = e^{t\mu + \frac{1}{2}t^2\sigma^2}$, and ~~first 2 central moments~~
 $E(X) = \mu, \text{Var}(X) = \sigma^2$.

Sp. case $\mu=0, \sigma^2=1, Z = \frac{X-\mu}{\sigma} \sim N(0,1)$, Z follows a standard normal distribution.

(can be shown using any of the 3 methods, i.e., distribution fn, mgf or Jacobian method)

Use notation $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, z \in \mathbb{R}$ for pdf, and

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z \phi(t) dt \quad \text{for cdf of } Z$$

Clearly, $\phi(z) = \phi(-z)$ and $\Phi(-z) = 1 - \Phi(z) \quad \forall z \in \mathbb{R}$

⑥ Normal approximation to a discrete r.v., say, $X \sim \text{Bin}(np, p)$

(works nicely for large n , and p not at the extremes, i.e., $p \neq 0$ or 1)

$$P(a \leq X \leq b) = P\left(\frac{a-np}{\sqrt{npq}} \leq \frac{X-np}{\sqrt{npq}} \leq \frac{b-np}{\sqrt{npq}}\right) \approx \Phi\left(\frac{b-np}{\sqrt{npq}}\right) - \Phi\left(\frac{a-np}{\sqrt{npq}}\right)$$

(by CLT, $Z = \frac{X-np}{\sqrt{npq}}$ converges in distrib to $N(0,1)$).

However, since we're considering a discrete r.v. with jump size 1, we need to use the continuity corrections for better approx, and

$$\text{hence, } P(a \leq X \leq b) \approx \Phi\left(\frac{b-np+1/2}{\sqrt{npq}}\right) - \Phi\left(\frac{a-np-1/2}{\sqrt{npq}}\right).$$

* ⑦ the p -dimensional random vector $X = (X_1, \dots, X_p)'$ follows p -variate multivariate normal $N(\mu, \Sigma)$ with pdf

$$f_X(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|} \exp. -\frac{1}{2}(x-\mu)' \Sigma^{-1}(x-\mu), \quad x \in \mathbb{R}^p, \mu \in \mathbb{R}^p, \Sigma \in \mathbb{R}^{p \times p}.$$

where $\mu_{px1} = (\mu_i), \mu_i = E(X_i), i=1(1)p$
 $\Sigma_{pxp} = ((\sigma_{ij})), \sigma_{ij} = \text{Cov}(X_i, X_j); i,j=1(1)p$

Check the pdf of bivariate normal pdf with $p=2$.

3. $X \sim \text{Gamma}(\alpha, \beta)$, α : shape, β : scale

with pdf $f_X(x) = \frac{1}{\Gamma(\alpha) \beta^\alpha} \exp(-x/\beta) x^{\alpha-1}$; $x > 0$; $\alpha, \beta > 0$

$M_X(t) = \frac{1}{(1-\beta t)^\alpha}$ exists if $t < \frac{1}{\beta}$.

r th order raw moment $E(X^r) = \frac{\Gamma(\alpha+r)}{\Gamma(\alpha)} \beta^r$

Spl. case $\alpha = 1$, $X \sim \text{Exp}(\beta)$ with pdf

$f_X(x) = \frac{1}{\beta} e^{-x/\beta}$, $x > 0$; $\beta > 0$

lack of memory property $P(X \geq r+s | X \geq s)$ does not depend on s .

Spl. case $\alpha = \frac{n}{2}$, $\beta = 2$, then

$X \sim \chi_n^2$ (same as $G(\frac{n}{2}, 2)$), for positive integer n

Common variants of the exponential distribution

4. $X \sim \text{Exp}(\mu, \beta)$; 2-parameter exponential distⁿ with location μ and scale β

pdf $f_X(x) = \frac{1}{\beta} \exp(-\frac{1}{\beta}(x-\mu))$, $x > \mu$ ($\mu \in \mathbb{R}$, $\beta > 0$) (support depends on the parameter, like the uniform distribution)

5. not to be confused with the Double Exponential distribution

$X \sim \text{DE}(\mu, \beta)$ with pdf

$f_X(x) = \frac{1}{2\beta} \exp\left\{-\frac{1}{2\beta}|x-\mu|\right\}$, $x \in \mathbb{R}$ (imp.), $\mu \in \mathbb{R}$, $\beta > 0$

6. $X \sim \text{Beta}(\alpha, \beta)$ with pdf

$f_X(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$, $0 < x < 1$; $\alpha, \beta > 0$

Spl. case $\alpha = \beta = 1$, $X \sim U(0, 1)$

7. $X \sim \text{Cauchy}(\mu, \sigma)$ (μ, σ) location, scale

$f_X(x) = \frac{\sigma}{\pi} \cdot \frac{1}{\sigma^2 + (x-\mu)^2}$, $x \in \mathbb{R}$; $\mu \in \mathbb{R}$, $\sigma > 0$

Mgf and hence moments (of any order) do not exist

but moments of order statistics exist under some restrictions

NOTE ① Exponential, 2-parameter exp., Weibull, log. normal, etc. are some distributions ~~which~~ fit for lifetime data & hence used effectively in Survival or Reliability analysis

② Double Exponential, Cauchy, t , mixture normal, etc. are some heavy-tailed distributions effective for financial analysis.